



Seasonal Trends in Animal Shelter Adoptions: A Strategic Analysis



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Data Overview



Source: [Austin, TX Animal Center Outcomes](#)

Dataset Size: Originally 173,775 rows, 12 columns

Scope: Focused on shelter Cats and Dogs with an outcome of "Adoption"

Data Cleaning:

- Parsed dates to extract Month/Year
- Filtered out incomplete years to ensure seasonal accuracy
- Standardized Seasons (Winter, Spring, Summer, Fall)





79,307

Rows analyzed after cleaning



45,245

Dog adoptions analyzed



34,062

Cat adoptions analyzed



Seasonal Adoptions Overlook



Winter

Dogs: 11,923
Cats: 7,189



Spring

Dogs: 10,931
Cats: 4,793



Summer

Dogs: 11,929
Cats: 12,528



Fall

Dogs: 10,462
Cats: 9,552





Research Questions



Question 1

Is there a significant seasonal dependency in adoption rates for shelter dogs?

Null Hypothesis

Adoption counts for Dogs are independent of the season.

Alternate Hypothesis

Adoption counts for Dogs vary significantly depending on the season.



Question 2

Is there a significant seasonal dependency in adoption rates for shelter cats?

Null Hypothesis

Adoption counts for Cats are independent of the season.

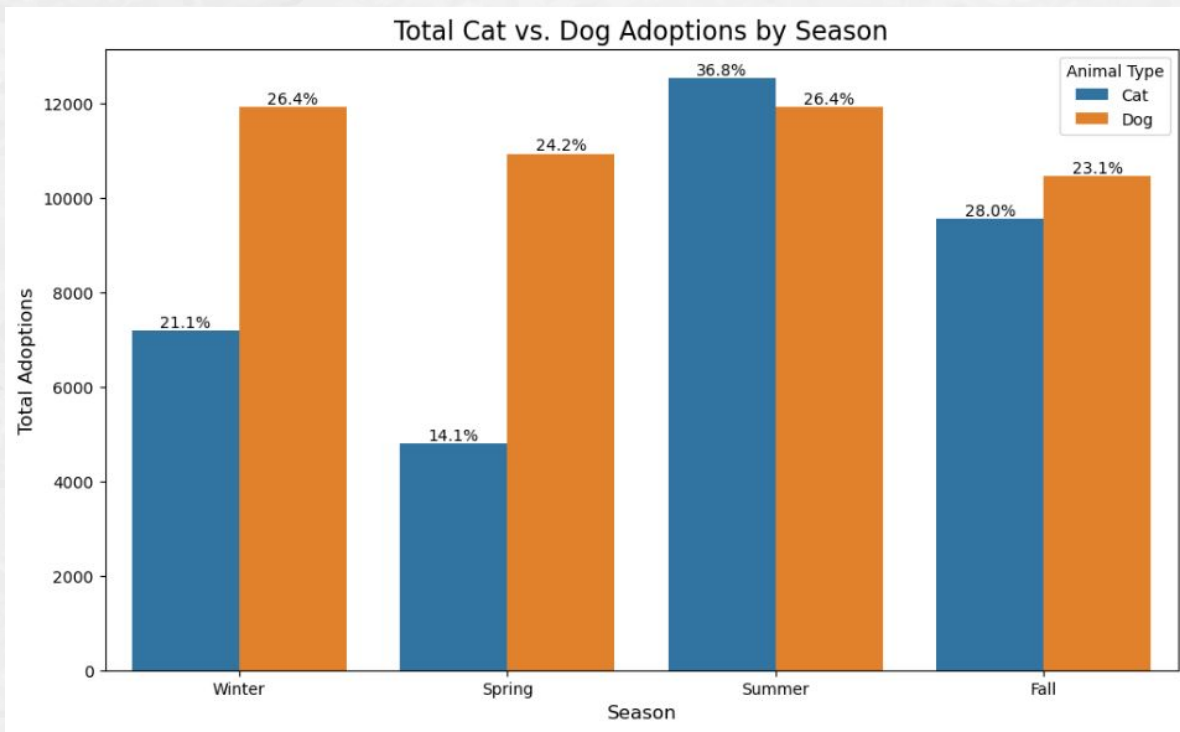
Alternate Hypothesis

Adoption counts for Cats vary significantly depending on the season.





Visual Analysis: Trends



Summer is the peak season for both animals.



Spring has the lowest adoption numbers for both.



Dogs seem to have a steady adoption rate, fluctuating ~2-3%



Cat adoptions decline following the Summer months.

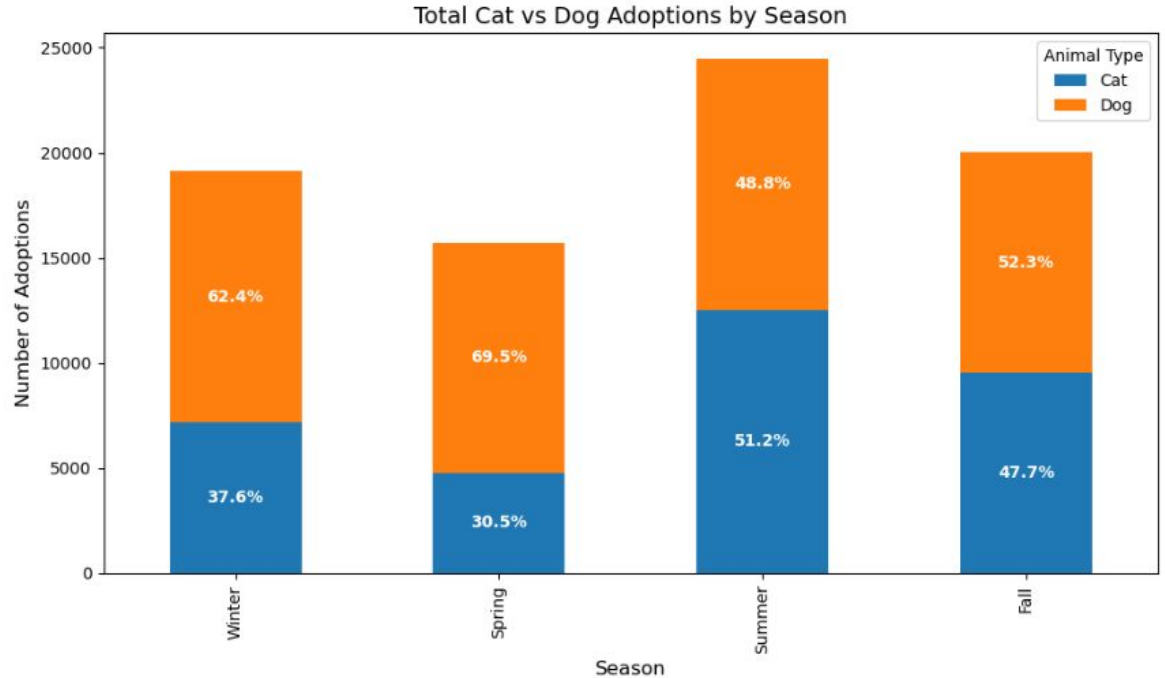
Visual Analysis: Proportions



This view maps total seasonal volume per species to aid in optimizing resource allocation.

Summer will require the most resources to manage the highest traffic volume.

Summer is the only season where cat adoptions surpasses dog adoptions.





Statistical Findings – Shelter Dogs

Chi Square Statistic
~143.37

Interpretation

We reject the null hypothesis.
Dog adoptions are statistically
dependent on the season.



**Chi-Square
Goodness-of-Fit
Test**

P-value
7.09e-31



Statistical Findings – Shelter Cats



**Chi-Square
Goodness-of-Fit
Test**

**P-value
0.0***

*value too small to compute

**Chi Square Statistic
~3850.7**

Interpretation

**We reject the null hypothesis.
Cat adoptions are statistically
dependent on the season.**



Conclusions

Cats:

The Summer volume surge seems to be driven by “Kitten Season,” the peak breeding period for felines, that trails into the Fall (Arnold 2023). Cat adoption counts trend downward significantly following this Summer peak. This can create a critical capacity crisis in Summer and Fall, leading to overcrowding and increased resource strain. Overall, shelter cat adoption rates significantly depend on the season.



Dogs:

Statistical analysis confirms significant seasonality in dog adoptions, with distinct peaks in Winter and Summer. Unlike the biological driver of 'Kitten Season,' dog trends can be driven by external factors: Winter peaks may reflect strays seeking shelter from the cold, while Summer spikes align with vacation-related owner surrenders (Reinert 2018). Further research can be conducted to analyze these external factors. Overall, dog adoption rates depend on the season and resources should be allocated efficiently.



Strategic Recommendations



Run aggressive marketing campaigns for Foster/Volunteer Applications and Cat Supply/Monetary Donations in the Late Spring/Early Summer

1

2

Conduct further analyses and tests to find statistically significant external factors that lead to dog intakes and outcomes that will aid in creating a plan to tackle dog adoptions.

3

Implement waived fees for Senior Animals during the Spring season to move harder to adopt animals out of the system before the Summer surge.



Thank You

Do you have any questions?

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CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon** and infographics & images by **Freepik**





References



GitHub link to Project:

<https://github.com/simsophie/Seasonal-Trends-in-Shelter-Cats-Dog-Adoption-Rates/>

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<https://www.cincinnati.com/story/news/local/northern-ky/2018/07/18/overcrowding-too-many-dogs-dropped-shelters/759247002/>